Predicting Showjumping Outcomes with Self-Supervised Video Representations

CS 131 Project Proposal: Type-Conditioned Approach Quality from Unlabeled Footage

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Problem Statement

Predicting the outcome of a horse's jump (clean, knockdown, refusal) is a motion analysis problem; stride rhythm, approach speed, and takeoff distance together determine success. However, predicting outcomes directly from video mixes two things up: how well the horse is approaching, and how hard the fence is. A vertical and an oxer of the same height are different problems, so a model trained on mixed fences may just learn fence type instead of approach quality. We fix the height at 1.50 m and predict approach quality given obstacle type (vertical vs. oxer). Our target is takeoff distance d, a standard equestrian quantity whose ideal value depends on jump type. Unlabeled footage is widely available but labels per jump are expensive, so we use self-supervised learning (SSL) to pretrain a video encoder on unlabeled clips, then fine-tune small heads on a labeled set. We test the following hypothesis: SSL-pretrained representations match from-scratch supervised accuracy with half the labels, and beat ImageNet pretraining with fewer still.

Technical Approach

Obstacle-aware preprocessing. YOLOv8 finds the horse and the fence then labels the fence type (vertical vs. oxer) from box shape and pole count. We recover d in meters using the standard 3.5 m pole as a scale reference, giving (type, d) alongside the raw video.

Method 1: Self-supervised video pretraining. We pretrain a small 3D-CNN encoder (R(2+1)D-18) on unlabeled clips with contrastive learning. Nearby frames are positives, frames from other clips are negatives. A second task (predicting the correct order of three shuffled sub-clips) is used to guide the encoder to learn motion instead of appearance. Future-frame embedding prediction is a stretch goal.

Method 2: Downstream fine-tuning. Two heads sit on top of the encoder (frozen and fine-tuned): a classifier for {clean, knockdown, refusal} given type, and a regressor for d. The regression target comes from video geometry.

Data

Unlabeled pretraining set: public 1.50 m showjumping footage from YouTube (FEI World Cup, Longines GCT, CSI 3* Grand Prix), cut into 2-second approach clips. Labeled set: hand-annotated outcomes and fence type across varied lighting and courses at 1.50 m.

Plan of Execution

1. **Week 1 (Data & preprocessing)**: scrape and segment footage; set up YOLOv8 detection and type classification; extract d from pole-length scaling; hand-label 100 clips.
2. **Week 2 (SSL pretraining)**: implement contrastive and temporal-order objectives; pretrain encoder; sanity-check with nearest-neighbor retrieval. **Fallback:** if retrieval doesn't cluster by approach phase, switch to a Kinetics-pretrained encoder.
3. **Week 3 (Downstream training & baselines)**: finish labeling 250 clips; train heads plus three baselines (from-scratch supervised, ImageNet 2D-CNN, type-only logistic regression).
4. **Week 4 (Analysis & writeup)**: sample-efficiency curves; ablations on pretext tasks and type conditioning; t-SNE colored by outcome, d, and type; final report and presentation.

Evaluation Strategy

Quantitative. (1) Outcome accuracy and macro-F1, broken down by type. (2) MAE of d regression in meters. (3) Sample-efficiency curves: accuracy vs. $n \in \{25, 50, 100, 200\}$, with SSL compared against from-scratch and ImageNet baselines. This is the main test of whether SSL cuts label requirements.

Qualitative. Failure-mode inspection (camera angle, occlusion, unusual courses) and t-SNE of pretrained embeddings, to see whether approach quality—and the vertical/oxer split—emerges without supervision.

Risk mitigation:. the Kinetics fallback handles a weak SSL encoder; geometric d regression still works even with noisy outcome labels; if 1.50 m footage is too scarce, we widen to 1.45–1.55 m and use pole-length calibration to absorb the small height differences. We will begin by analysing images before moving to video as well to mitigate risks of not completing the full project.